

Chapter 16

Magnetostatics in materials

16.1 Magnetization and the field $\vec{H}$

In a material, the molecules can carry magnetic dipole moments $\vec{m}_{\text{mol}}$. Adding their dipole vector potentials to the contribution of the currents, we have

$$\vec{A} = \frac{1}{c} \int \frac{\vec{J}(\vec{x}')}{|\vec{x} - \vec{x}'|} dV' + \sum_i \frac{\vec{m}_{\text{mol}} \times (\vec{x} - \vec{x}_i)}{|\vec{x} - \vec{x}_i|^3} \quad (16.1)$$

Introducing the magnetization $\vec{M}$ of the material – the magnetic dipole moment per unit volume – the sum over molecules becomes

$$\sum_i \frac{\vec{m}_{\text{mol}} \times (\vec{x} - \vec{x}_i)}{|\vec{x} - \vec{x}_i|^3} = \int \vec{M}(\vec{x}') \times \vec{\nabla}' \frac{1}{|\vec{x} - \vec{x}'|} dV' \quad (16.2)$$

We then write

$$\int \vec{M}(\vec{x}') \times \vec{\nabla}' \frac{1}{|\vec{x} - \vec{x}'|} dV' = - \int \vec{\nabla}' \times \frac{\vec{M}(\vec{x}')}{|\vec{x} - \vec{x}'|} dV' + \int \frac{\vec{\nabla}' \times \vec{M}(\vec{x}')}{|\vec{x} - \vec{x}'|} dV' \quad (16.3)$$

The first term on the right-hand side is a surface integral that vanishes for a localized material

$$\int \vec{\nabla}' \times \frac{\vec{M}(\vec{x}')}{|\vec{x} - \vec{x}'|} dV' = \oint \frac{d\vec{a}' \times \vec{M}}{|\vec{x} - \vec{x}'|} \rightarrow 0 \quad (16.4)$$

We are then left with

$$\vec{A}(\vec{x}) = \frac{1}{c} \int \frac{\vec{J}(\vec{x}') + c \vec{\nabla}' \times \vec{M}(\vec{x}')}{|\vec{x} - \vec{x}'|} dV' \quad (16.5)$$

which has the form of the magnetostatic solution with the current density shifted by the *magnetization current*

$$\vec{J}_M \equiv c \vec{\nabla} \times \vec{M} \quad (16.6)$$

Ampère's law correspondingly picks up this additional source

$$\begin{aligned} \vec{\nabla} \times \vec{B} &= \frac{4\pi}{c} (\vec{J} + \vec{J}_M) = \frac{4\pi}{c} (\vec{J} + c \vec{\nabla} \times \vec{M}) \\ \Rightarrow \vec{\nabla} \times (\vec{B} - 4\pi \vec{M}) &= \frac{4\pi}{c} \vec{J} \end{aligned} \quad (16.7)$$

We define

$$\vec{H} \equiv \vec{B} - 4\pi \vec{M} \quad (16.8)$$

called the *magnetic field intensity*: it is sourced by the free currents only.

16.2 Linear materials

For linear materials, the magnetization responds linearly to the field

$$\vec{M} = \chi_m \vec{H} \quad (16.9)$$

where χ_m is the magnetic susceptibility. Inverting (16.8) then gives

$$\vec{B} = \vec{H} + 4\pi \vec{M} = (1 + 4\pi \chi_m) \vec{H} \equiv \mu \vec{H} \quad (16.10)$$

where μ is called the permeability. There are three cases:

$$\mu < 1 \quad (\chi_m < 0) \rightarrow \text{diamagnetism} \quad (16.11)$$

as for water, with $\chi_m \simeq -1.2 \times 10^{-5}$;

$$\mu > 1 \quad (\chi_m > 0) \rightarrow \text{paramagnetism} \quad (16.12)$$

which is temperature dependent, as for O_2 with $\chi_m \simeq 1.8 \times 10^{-6}$ [?]; and

$$\mu \gg 1 \rightarrow \text{ferromagnetism} \quad (16.13)$$

as for Fe, with $\chi_m \sim 10^6$. Ferromagnets are often non-linear, $\mu(H)$, exhibiting hysteresis as depicted in the Figure: at $H = 0$ the material retains a permanent magnetization.

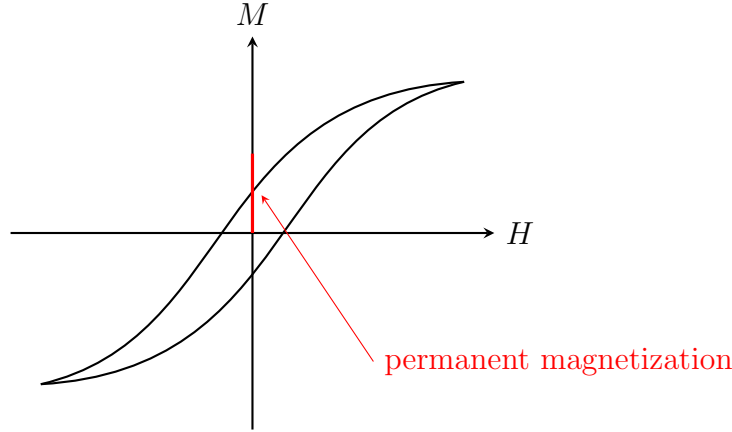


Figure 16.1: Hysteresis in a ferromagnet: the relation $M(H)$ is multivalued, and at $H = 0$ a permanent magnetization remains (highlighted segment).

16.3 Summary and boundary conditions

To summarize, the magnetostatic equations in a material read

$$\vec{\nabla} \cdot \vec{B} = 0 \quad , \quad \vec{\nabla} \times \vec{H} = \frac{4\pi}{c} \vec{J} \quad (16.14)$$

with $\vec{J}$ the free current. Across an interface between two materials with normal $\hat{n}$, these imply

$$(\vec{B}_2 - \vec{B}_1) \cdot \hat{n} = 0 \Rightarrow \mu_1 H_{1n} = \mu_2 H_{2n} \quad (16.15)$$

and

$$(\vec{H}_2 - \vec{H}_1) \times \hat{n} = \frac{4\pi}{c} \vec{K} \rightarrow 0 \quad \text{for no free surface currents} \quad (16.16)$$

16.4 Energy

Consider the work needed to build up the currents. Changing the flux Φ through a circuit carrying current I induces a back EMF $\mathcal{E}$, against which we do work

$$\Delta W = I \Delta \Phi = I \int \delta \vec{B} \cdot d\vec{a} \quad (16.17)$$

since $\Delta W = -I \mathcal{E} \Delta t = I \Delta \Phi$. We then write

$$\begin{aligned}
 \Delta W &= I \int (\vec{\nabla} \times \delta \vec{A}) \cdot d\vec{a} = I \oint \delta \vec{A} \cdot d\vec{s} \rightarrow \int \delta \vec{A} \cdot \vec{J} dV \\
 &= \int \delta \vec{A} \cdot (\vec{\nabla} \times \vec{H}) dV \\
 &= \int (\vec{H} \cdot (\vec{\nabla} \times \delta \vec{A}) + \vec{\nabla} \cdot (\vec{H} \times \delta \vec{A})) dV \\
 &= \int \vec{H} \cdot \delta \vec{B} dV = \int \frac{1}{2} \delta (\vec{H} \cdot \vec{B}) dV
 \end{aligned} \tag{16.18}$$

where in the first line we generalized from a single circuit to a current distribution by $I d\vec{s} \rightarrow \vec{J} dV$, in the third line the total-divergence term drops since the fields are localized, and in the last step we assumed a linear material. Hence

$$W = \frac{1}{2} \int \vec{H} \cdot \vec{B} dV = \frac{1}{2} \int \vec{J} \cdot \vec{A} dV \tag{16.19}$$

If instead a material is introduced into a pre-existing field $\vec{B}_0 = \mu_0 \vec{H}_0$ maintained by fixed sources, the change in energy is

$$\begin{aligned}
 \Delta W &= \frac{1}{2} \int (\vec{B} \cdot \vec{H}_0 - \vec{H} \cdot \vec{B}_0) dV = \frac{1}{2} \int (\mu - \mu_0) \vec{H} \cdot \vec{H}_0 dV \\
 &= \frac{1}{2} \int \vec{M} \cdot \vec{B}_0 dV
 \end{aligned} \tag{16.20}$$

the last equality holding if there are no free currents.

For two circuits i, j , the energy

$$W = \int dV dV' \frac{\vec{J}(\vec{x}) \cdot \vec{J}(\vec{x}')}{|\vec{x} - \vec{x}'|} = I_i I_j M_{ij} \tag{16.21}$$

defines the *mutual inductance* M_{ij} . For a single circuit, the *self inductance* L is defined by

$$W = \frac{1}{2} L I^2 \tag{16.22}$$

Altogether, for a collection of circuits

$$W_{\text{tot}} = \sum_i \frac{1}{2} L_i I_i^2 + \sum_i \sum_{j>i} M_{ij} I_i I_j \tag{16.23}$$